

Final Exam Software Change Implementation Report

CR-006: Last Login Tracking and Display for Student Accounts

1. Change Request Details

Field	Details
Change Request ID	CR033
Title	Last Login Tracking and Display for Student Accounts
Priority	Medium
Difficulty	Low
Change Type	Database, Backend and Frontend Enhancement
Affected Version	Student Management System v1.0 baseline
Dependency	Existing authentication/login flow and existing GET /api/students/ endpoint

2. Business Requirement

Administrators currently can view student account and profile information, but the student list does not show when a student last logged in. The system should record the date and time of a successful student login and display this information in the administrator Students table. This allows administrators to understand recent account activity and identify students who have never logged in.

3. Current Behaviour

- The login process validates the username and password and creates a JWT token after successful authentication.
- The users table stores account and role information, but no last login timestamp is displayed in the Students list.
- The administrator Students table displays Registration No., Name, Email, Phone, and Actions.
- Administrators cannot currently see the date and time of a student's most recent successful login.

4. Requested Change

Add a Last Login field to track and display the most recent successful login time for each student. The implementation must:

- Add a last_login DATETIME field to the users table.
- Update last_login only after the username and password have been successfully verified.
- Store the current database date and time using NOW().
- Return last_login with student records through the existing student API.
- Display a Last Login column in the administrator Students table.
- Format the stored timestamp into a readable date and time in the frontend.
- Display “Never logged in” when a student has no recorded login.
- Preserve existing student Add, Edit, Delete, authentication, and authorization functionality.

5. Scope

In Scope

- Users table last_login column.
- Successful login timestamp update.
- Student API query update to return last_login.
- Last Login column in the administrator Students table.
- Readable date/time formatting.
- Never logged in state for accounts without a login record.
- Preservation of existing CRUD and authentication functionality.

Out of Scope

- Password reset functionality.
- Changing JWT expiration behaviour.
- New authentication methods.
- Login history or multiple previous login records.
- Pagination or sorting changes.
- Changes to existing administrator authorization rules.

6. Affected Files

- database/schema.sql — add the last_login column to the users table.
- backend/models/userModel.js — update user queries and add updateLastLogin().
- backend/controllers/authController.js — update last_login after successful password verification.
- backend/models/studentModel.js — include users.last_login in student queries using a JOIN.
- frontend/src/pages/DashboardPage.jsx — display the Last Login column and format the timestamp.

7. Functional & Security Requirements

- FR-001: The users table shall store a nullable last_login timestamp.
- FR-002: last_login shall be updated only after successful username and password verification.
- FR-003: The existing login response and JWT authentication flow shall continue to work.
- FR-004: Student records returned by the existing student API shall include last_login.
- FR-005: The administrator Students table shall display Last Login.
- FR-006: A student without a previous successful login shall display “Never logged in”.
- FR-007: Existing Edit and Delete operations shall continue to use the correct student_id.
- SEC-001: Passwords shall continue to be verified using bcrypt.
- SEC-002: The change shall not expose password hashes or other additional sensitive authentication data.
- SEC-003: Existing administrator authentication and authorization shall remain unchanged.

8. Implementation Summary

The Last Login feature was implemented across the database, backend authentication flow, student data retrieval, and administrator frontend. A nullable last_login column was added to the users table. After bcrypt password verification succeeds, updateLastLogin() updates the user's timestamp. The student model joins students with users so the last login timestamp is returned to the frontend. DashboardPage.jsx displays the value in a new Last Login column and shows “Never logged in” when no login has been recorded.

9. Test Cases and Results

Test Case	Expected Result	Result	Status
Successful student login	last_login is updated with current date/time.	Timestamp updated.	PASS
Incorrect password	last_login is not updated.	Login rejected; timestamp unchanged.	PASS
Student list load	Last Login value is returned with student data.	Value displayed.	PASS
Student with no previous login	Display “Never logged in”.	Fallback message displayed.	PASS
Existing Edit action	Correct student can still be edited.	Existing behaviour preserved.	PASS
Existing Delete action	Correct student can still be deleted.	Existing behaviour preserved.	PASS

10. Acceptance Criteria Verification

- AC-001: Successful login records the student's latest login date and time — VERIFIED.
- AC-002: Incorrect login credentials do not update the last login timestamp — VERIFIED.
- AC-003: The Students table displays a Last Login column — VERIFIED.
- AC-004: Students with no recorded login display “Never logged in” — VERIFIED.
- AC-005: Existing student management CRUD operations continue to work — VERIFIED.
- AC-006: Existing authentication and authorization behaviour remains unchanged — VERIFIED.

11. Expected Outcome

Administrators can now see when each student last successfully logged in directly from the Students table. This improves account activity visibility while preserving the existing student management, authentication, authorization, and CRUD functionality.

12. Screenshots / Evidence

- Screenshot 1: Students table showing the new Last Login column.
- Screenshot 2: Successful student login followed by the updated Last Login value.
- Screenshot 3: Student account showing “Never logged in” before the first successful login.
- Screenshot 4: Existing Edit/Delete actions working after the change.

13. Final Status and Outstanding Issues

Final Status: IMPLEMENTED

Outstanding Issues: None identified for the requested Last Login feature.